# Smart Insole for Foot Ulcer Prediction

## **Code:** CS9

## Team Members:


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## Abstract:


This project presents the development of an intelligent healthcare system that integrates Artificial Intelligence, Embedded Systems, and Mobile Application technologies to support early disease prediction and patient monitoring. The core of the system is a Smart Insole designed to measure foot pressure and temperature in real time, enabling early detection of conditions associated with diabetic foot ulcers using machine learning models.
In addition to ulcer prediction, the system includes multiple AI-powered modules such as diabetic foot ulcer image classification with severity grading, stroke risk prediction based on patient input data, and a general health assessment module capable of predicting multiple healthcare conditions.
The proposed platform also provides several supporting features, including personalized nutrition planning, medical report analysis, future health condition simulation, result explanation, and an AI-based chatbot assistant to enhance user interaction and understanding.
By combining hardware sensing, artificial intelligence, and mobile healthcare technologies, the proposed system provides a comprehensive and scalable healthcare solution aimed at improving early diagnosis, preventive healthcare, and patient quality of life.